

JUNGIN KIM

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SUMMARY

- 4 years of research experience in NLP (2 years in a master program, 2 years as research intern)
- 1 year of hands-on experience in data analysis and machine learning within industry environments.
- Experience in academic research and patent development in NLP (5 papers and 2 patents).

RESEARCH INTERESTS

- **AI Safety** – Watermarked generation for identifying LLM-generated content and tracing content origin.
- **Hate Speech Detection** – Detection and evaluation of explicit and implicit harmful language.
- **Retrieval Systems** – Design and optimization of LLM-based generation and retrieval systems.
- **Recommender Systems** – Personalized recommendation models based on user behavior logs.

EDUCATION

Yonsei University, Sep 2023 - Aug 2025

M.S. in Computer Science, Theory of Computation Lab

GPA: 4.01/4.30 | Advisor: Prof. Yo-Sub Han | Research Area: NL/Code Generation, LLM Watermarking

Sangmyung University, Mar 2017 - Feb 2022

B.S. in Computer Software, Network Data Science Lab

GPA: 3.92/4.50 | Advisor: Prof. Hyunchul Kim | Research Area: Hate Text Classification

WORK EXPERIENCES

Research Assistant, Aug 2023 - Present

Theory of Computation Lab (Advisor: Prof. Yo-Sub Han) , Yonsei University

- Conducted large-scale experiments on 399+ code generation tasks, improving execution accuracy by 7.57%p while maintaining high AUROC and reducing detection latency by 86%.
- Constructed a 28K-sentence dialectal hate speech dataset and improved span-level detection accuracy by 36.70%p using curriculum learning, enabling more effective toxicity mitigation (+13.29%p).
- Built and human-annotated a large-scale code dataset (9,800 samples) and leveraged it to design semi-supervised experiments, achieving over 60% gains in accuracy and F1 through data augmentation and co-training.
- Improved cross-dataset generalization in implicit hate speech detection using contrastive learning (+1.43 F1).

Full time Data Analyst, Mar 2022 – Apr 2023

Data Team, Alookso, Inc.

- Developed and deployed a hybrid personalized content recommendation system combining matrix factorization and KoBERT-based content similarity, serving main-page recommendations over 50,000 users.
- Developed and deployed a new algorithm that embeds users based on survey responses and visualizes user similarity via dimensionality reduction, enabling personalized user recommendations in the service.

Contract Research Assistant, Aug 2020 – Nov 2020

Statistical Artificial Intelligence Lab (Advisor: Prof. Jaesik Choi), KAIST

- Built a building price prediction model using 10 unstructured appraisal reports PDFs, designing a rule-based feature extractor grounded in professional appraisal criteria to convert into structured valuation features.
- Implemented an ensemble model combining cost, comparison, and income approaches, improving valuation accuracy over single-method baselines.

Research Intern,

Mar 2019 – Jul 2020

Network Data Science Lab (Advisor: Prof. Hyunchul Kim), Sangmyung University

- Hands-on experience in manually annotating 600+ hate speech samples for attention-based error analysis, defining span-level labels to diagnose model failure cases and interpretation errors.
- Trained a Hierarchical Attention Network achieving 89% accuracy, visualized attention weights for explainability, and integrated the model into a Telegram bot for real-time hate speech span detection.

Data Analyst Intern,

Aug 2017 – Aug 2017

Game Development Team (3on3 FreeStyle), Joycity, Inc.

- Identified key drivers of user churn through analysis of sales, and community data, and proposed summer event and character balance changes to improve retention.

PUBLICATIONS

“K-ToPiC: Korean Dialect Toxicity Polishing with Curriculum Learning-based Span Detection”

Jungin Kim, Soohan Lim, Hyeseon Ahn, Sang-Min Choi, Yo-Sub Han

In Proceedings of the Korea Computer Congress: KCC 2025. ([paper](#))

“Marking Code Without Breaking It: Code Watermarking for Detecting LLM-Generated Code”

Jungin Kim, Shinwoo Park, Yo-Sub Han

Master’s Thesis, arXiv preprint arXiv:2502.18851 [Under Review], 2025.([paper](#))

“TCProF: Time-Complexity Prediction SSL Framework”

Joonghyuk Hahn, Hyeseon Ahn, **Jungin Kim**, Soohan Lim, Yo-Sub Han

In the Association for Computational Linguistics: NAACL 2025.([paper](#))

“CodeComplex: A Time-Complexity Dataset for Bilingual Source Codes”

Seung-Yeop Baik, Joonghyuk Hahn, **Jungin Kim**, Mingi Jeon, Yo-Sub Han, Sang-Ki Ko

In Findings of the Association for Empirical Methods in Natural Language Processing: EMNLP 2025.([paper](#))

“SharedCon: Implicit hate speech detection using shared semantics”

Hyeseon Ahn, Youngwook Kim, **Jungin Kim**, Yo-Sub Han

In Findings of the Association for Computational Linguistics: ACL 2024.([paper](#))

PATENTS

Code Watermarking Method and Apparatus that Preserves Syntax Application

No. 10-2025-0021267 | Filed: Feb 19, 2025

Code Complexity Prediction Method and Apparatus Using Semi-supervised Learning Application

No. 10-2024-0150149 | Filed: Oct 29, 2024

AWARDS & HONORS

- ADsP (Associate Data Analyst Certificate), Korea Data Agency Apr 2023
- Completed AIFTEL AI Bootcamp, Modu Research Institute Dec 2021 - Feb 2022
- Completed, Recommender System Study, Modu Research Institute Aug 2021
- Graduated, Alice AI Track (1st Cohort) Dec 2020 - Jun 2021
- Awarded First Place, University Capstone Design Competition Sep 2020 - Nov 2020
- Awarded First Place, SK Future Industry Technology Startup Competition May 2017

LANGUAGES, SKILLS

- Languages: Korean, English, TOEIC 860, OPIC IM2
- Programming Skills: Python, C, R, HTML, CSS, PyTorch, SQL, Scikit-learn, Git, Redash